not be determined experimentally, nor should any change in setting the valves be made unless the effect of the change is known in advance. It is therefore important that the different parts should be made and set correctly from the beginning, and there will then be no need for changes when the original dimensions are maintained. The difference in this gear for outside and inside admission valves must be considered in setting the eccentric crank and as the forward motion of the engine should preferably be taken from the lower end of the link when the eccentric crank will *follow* the main crank for *inside* admission valve (see Figs. 10 and 10a), and *lead* the main crank for *outside* admission valve. (See Fig. 11). The connecting point of the radius bar to the combination lever is *above* that of the valve for stem connection for *inside* admission and *below* the valve stem connection for *outside* admission valves. (See Figs. 10, 10a and 11). The desired maximum cut-off, lead and valve travel determines the size of the lap, and thereby the lap and lead motion ob-

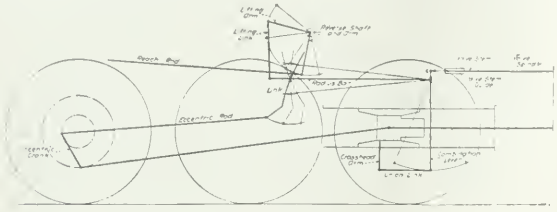


Fig. 10.

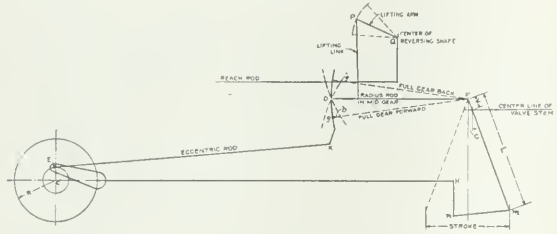


Fig. 10a.

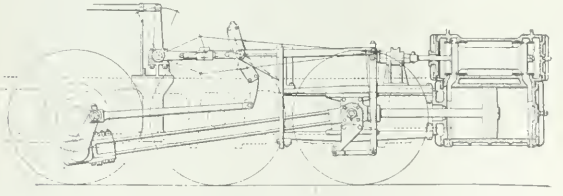


Fig. 11.

tained by the corresponding proportioning of the combination lever, and is found in the following formula:

R: C = L: V.

R = radius of the main crank.
C = lap and lead (one side).
L = distance between radius bar and crosshead connection (from F to M, Fig. 10a) on the combination lever.
V = distance between the radius bar and valve stem connections.

The length of the combination lever must be taken to suit the conditions under consideration in each case, so that the angle through which it oscillates will not exceed 60 deg., but less is preferable. The required horizontal movement or travel of the connecting point F of the radius bar to the combination lever for a given maximum valve travel must now be ascertained and is found by the following formula, in which R and C are the same as above, namely:

R = radius of main crank.
C = lap and lead.
a = half of the travel of the valve.
b = half the travel of point F.

$$b = \frac{R\sqrt{a^2 - C^2}}{R + C}$$ for outside admission, and

$$b = \frac{R\sqrt{a^2 - C^2}}{R - C}$$ for inside admission valve.

These may also be laid out graphically as per Fig. 12 for outside and Fig. 13 for inside admission valves by drawing a circle with S as a center and a as radius (shown dotted in the figures). Lay out crank radius R to the left from S = Sd and the lap and lead dimension C = Se on same side of S for inside admission (Fig. 13), and on opposite side of S for outside admission (Fig. 12). Draw ef and Sh perpendicular to Sd, when f becomes the intersecting point between the valve travel circle and the line ef. Draw the line df, and where this intersects the line Sh, which point we will call h (found by extending df in Fig. 13); Sh is then the desired dimension b in the formula, or one-half the required movement of point F, the total of which is represented by the full-drawn circle in the figures.

This is a most important function of the gear upon which prac-

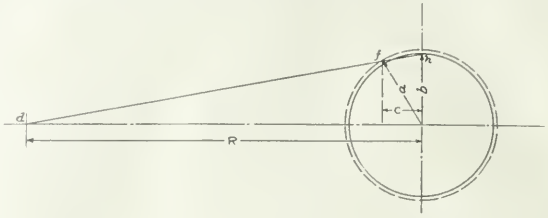


Fig. 12.

tically all the others depend, and is rather complicated to find by plotting. With a correct suspension of the link block it will have the same horizontal movement as the point F, and by limiting the angle of the swing of the link to 45 deg. as a maximum we get the rise or depression of the link block on either side of the link fulcrum the distance $Og = \frac{b^*}{\tan. d}$, (Fig. 10a) where O is the link fulcrum, d = one-half of the swing of the link in degrees, and b = half the travel of point f in the previus formula.

The vertical location of the link fulcrum O should be, when practicable, on a line drawn through point F parallel with the valve stem, and the eccentric rod connecting pin K to the link should be as nearly as practicable on the same level as the main axle in order to minimize the effect of the vertical play of the axle on the valve events, but on large engines it may be found necessary to lower fulcrum O and raise connection K to avoid excessive throw of the eccentric crank.

In locating the longitudinal position of the link fulcrum consideration must be given to the lengths of the eccentric and radius bars so that both may be of approximately the same length. When these lengths fall below three and one-half times the total vertical sweep

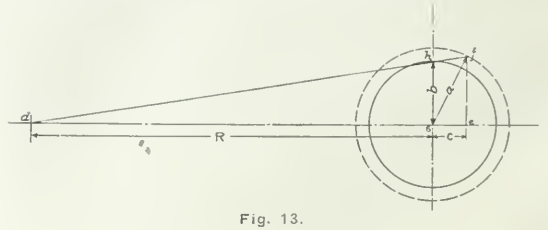


Fig. 13.



Fig. 14.

of the link block the radius bar should be favored in preference to the eccentric rod. The exact position of the eccentric crank must be plotted as well as the longitudinal location of point K. The former must bear such relation to the main crank that it brings the link in its middle position, when the main crank is on either of its dead centers and the connecting point K must be so located that it swings the link in the required angle *d* on either side of the middle position of the link; that is, in other words, the point K should be

*As no suspension gives a perfectly equal drop of the block in both link positions this formula is only approximate.

so located on the curve it must follow with fulcrum O as a center that its deviation from the tangent of the eccentric rod to this curve is such that it, as near as practicable, compensates for the irregularities brought about by the angularities of the main and eccentric rods which in ordinary cases brings it from 2 in. to 5 in. in the rear of the tangent to the link drawn through the fulcrum O.

The locus of the suspension point of the radius bar lifting link must also be plotted so that the link block is at the same point of the link in its extreme positions at all cut-offs. This locus is a curve

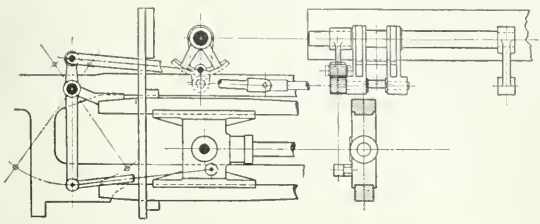


Fig. 15—Allfree Valve Gear.

with its center in the vicinity of the point F when in its mid-gear position. It would be impracticable, however, to have a lift-arm of this length, and a curve of smaller radius must be substituted and so applied that it intersects with the former curve at points giving the least possible distortion to the motion favoring the position of the link block in which it is mostly used in service.

The sliding lifter shown in Fig. 10 meets these conditions better than any other method of suspension, but, due to wheel arrangements of various designs of engines, this is not always applicable, but must be substituted by swinging lifters as per Figs. 10a and 11, which, when properly plotted, gives for all practicable purposes equally good results.

The vertical height of lower connection *m* of the combination lever in relation to the crosshead connection has a slight influence on the port opening, and should, therefore, in the center position of the lever, be about in the same level as the crosshead connecting point *n* (see Fig. 11).

In Fig. 14 is the motion of the valve graphically represented by

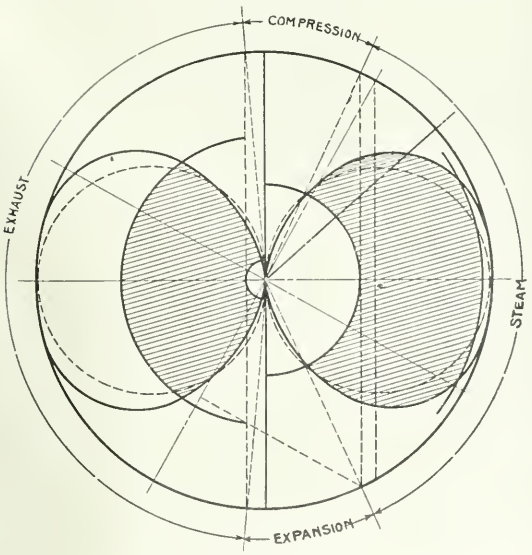


Fig. 16—Zeuner Diagram Applied to Allfree Valve Motion Attachment.

the Long diagram at different cut-offs where the horizontal lines represent the port opening edges in the cylinder face and the curves of the steam inlet edges of the valve, showing the opening and cut-off points where the latter intersect the former and the port openings at the various points of the stroke are measured by the height of these curves over and under the opening edges of the ports at both ends of the valve, respectively.

It will be noticed that these ellipses are slightly flattened on one side, which is caused by the slower lineal motion imparted to the valve relative to the angular motion when the eccentric passes its back center, compared with that of the front center, due to the angularity of the eccentric rod, and is more marked the shorter the rod. Fully symmetrical ellipses are not obtainable, as this would require the eccentric and main rods to be of infinite length, but this angularity, however, is of but little detriment to the distribution of the steam as long as the relations between the lengths of the eccentric rod and the throw of the eccentric is not less than the given limitations, and is present in all kinds of continuous valve motions derived from uniformly rotating cranks or eccentrics.

General Notes for Adjusting Walschaert Gear.—1. Ascertain by the following method the position of the eccentric crank: Mark the position of the link relative to its middle position on both of the dead centers of the main crank. If the position of the link is the same in both cases the eccentric crank position is correct, if not the eccentric crank should be lifted until this occurs or as near so as possible.

2. After the eccentric crank has been correctly set the eccentric rod should be lengthened or shortened as may be required to bring the link in its middle position, so that the link block can be moved from its extreme forward to its extreme backward position without imparting any motion to the valve. It may be noted that the link position may be observed by the usual tram marks on the valve stem, or direct by marks on the link pin as may be found most convenient with the link blocks in full gear, preferably ahead.

3. The difference between the two positions of the valve on the forward and back centers of the engine is the lap and lead doubled; it is the same in any position of the link block and cannot be changed by changing the leverage relations of the combination lever.

4. The tram marks of the opening moments at both ends of the

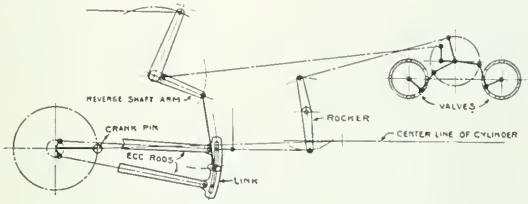


Fig. 17—Young's Rotary Valve Motion.

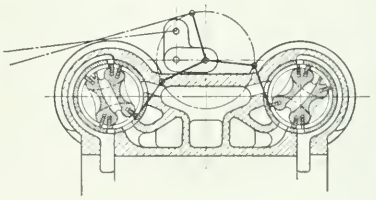


Fig. 18—Section Through Valves and Cylinders.

valve should be marked on the valve stem and the latter lengthened or shortened until equal leads at both ends are obtained.

5. Within certain limits this lengthening or shortening may be made on the radius bar, if it should prove more convenient, but it is desirable that its length should be so nearly equal to the radius of the link that no apparent change in the lead should occur in moving the link blocks as stated in note No. 2.

6. The lead may be increased by reducing the lap and the cut-off points will then be slightly advanced. Increasing the lap produces the opposite effect on the cut-off and reduces the lead the same amount. With good judgment these quantities may be varied to offset the irregularities inherent in transforming rotary into lineal motions.

7. The valve events are to a great extent dependent on the location of the suspension point of lifter of the rear end of the radius bar, when swinging lifter is used, which requires that this point should be properly laid out by careful plotting, or, if convenient, it is preferably determined by a model, as irregularities due to incorrect locus of this point cannot be corrected by the other parts of the gear without more or less distortion of same. When this point is so fixed that a change of same is impracticable it may be better, however, to modify other elements if thereby the motion in general can be improved.

The chief point of difference between the Walschaert and Stephenson gear, when both are in proper condition, is, as previously stated, that the former gives to the valve a constant lead at all cut-offs, whereas the latter produces an increase of lead by linking up the engine and becomes excessive at short cut-offs. This very point has been the subject for much controversy, and has probably done more than anything else to retard the progress of the use of Walschaert gear; as it has been argued that in full gear, when the speed generally is slow, only small lead is needed, but at higher speed more lead is required, which is accomplished by the Stephenson motion.

though this admittedly becomes excessive at early cut-offs, causing considerable compression and preadmission detrimental both to maintenance and to smooth running, and, in fact, to some degree counteracts the work done by the steam on the driving side of the piston, which thereby also affects the speed of the engine.

It was generally discovered that the required lead for short cut-off and high speed was of no practical detriment to the working of the engine in full gear, as the preadmission at that point is disappearingly small. The proper amount of lead, however, is dependent somewhat on the service, and the port opening becomes larger with a larger lead, or, in other words, when all other conditions are equal in a Stephenson or Walschaert gear the openings differ by the same amount as the lead, so that $^1/_{16}$ in. more lead gives $^1/_{16}$ in. wider port opening; but it is hardly advisable to make this over $^1/_4$ in. or $^5/_{16}$ in. as a maximum, as the advantage of any additional port opening by means of a larger lead is more than offset by the increase in compression and preadmission the larger lead would bring about at early cut-offs, and would do no good in the latter cut-offs, even if it does not do any harm.

There is no fundamental reason that the Walschaert gear should produce any economy in steam consumption over the Stephenson motion when both are in the best conditions, but an advantage in this respect comes to the former by the fact that it remains in its good condition if once made so, from one shopping to another, and is, therefore, on an average more economical both in steam consumption and maintenance of the gear than the latter. The accessibility for attention is a great point of undisputed advantage of the Walschaert gear, which is also highly appreciated by the enginemen and attendants.

It will be borne out in the course of time that the bracing between the frames permitted by the Walschaert gear will bring about a considerable reduction in the maintenance expenses by the less wear and tear this additional rigidity will impart to the entire engine.

Helmholtz Modification.—Among the various modifications of the Walschaert gear the one made by Helmholtz is probably of some advantage. This modification consists in making the link straight, and the radius bar is connected to the lifting link instead of the link block. The curving of the link is compensated for by the reversing shaft or lifting-arm fulcrum being located in a given position above the link, so that the locus of the suspension point of the lifting link forms an arc of a circle with its chord perpendicular to the center line of the radius bar in its center position. The radius of this arc bears the same relation to the length of the radius bar as the distance of the radius bar connection above the link block bears to the length of the lifting link, which results in that this connection is moving in an arc with a radius of the length of the radius bar and the same motion of the valve is obtained as in the direct Walschaert gear.

Two advantages may be claimed for this modification, of which one is the straight link being simpler to make than the curved one, and the other is that on large piston valve engines with inside admission the link fulcrum can be lowered by the amount the radius bar connection falls over the link block, whereby the eccentric rod connection can be brought closer to the center line of the axle with less length of link and eccentric throw. It has, however, the disadvantage that there is little choice in the location of the reversing shaft or lifting-arm fulcrum, a proper position for which is hardly obtainable on all types of engines and admits of no other method of lifting the radius bar in linking up or reversing the engine.

Allfree-Hubbell Gear Attachment.—The Allfree contrivance is an attachment to the ordinary Stephenson motion by which the valve is given a symmetrically irregular motion, causing it to open and close quickly, and is of special importance at high speed and early cut-off. The effect of this movement is about the same as what would be accomplished with a valve with extraordinarily long valve travel, but with a somewhat delayed exhaust and compression. The general construction of this device is shown in Fig. 15, and consists in substituting for the knuckle pin joining the valve stem to the rocker arm a small crank shaft, to which crank the valve stem is connected and the shaft left free to revolve in the rocker arm bearing. To this shaft a pinion is keyed engaging a toothed wheel sector oscillating on the rocker shaft by means of a lever connection from the crosshead. This gives a composite motion to the valve, namely, one direct from the eccentrics and one from the crosshead transmitted by the rotation of the crank on the knuckle pin shaft in such a way that the two motions coincide at the opening and closing moments, but are in opposition at the extreme travel of the valve, when the latter is nearly at a standstill, while the main crank passes through a comparatively large angle with a uniformly open port, and the closing of the valve is rapidly accelerated in the same manner as the opening, causing a quick and sharp cut-off. Fig. 16 gives an approximate illustration of the motion of the valve relative to the ordinary circular motions represented by a Zeuner diagram. The exhaust and compression are similarly affected by this alternating, accelerating and retarding motion of the valve, delaying the exhaust and compression even at early cut-offs. The advantage is, as already stated, greatest at high speed when a relatively high average pressure is obtained, which again is dependent on the capacity of the boiler. It is quite complicated to attach, especially on certain types of engines when the driving wheels are straddled by the guide yoke and thereby limiting the space required for the rocker arms and bearings.

Young Valve Arrangement.—This gear consists chiefly in the application of the Corliss valves to the locomotive engine with one valve both for the steam inlet and the exhaust at each end of the cylinder. Each valve is provided with double admission and exhaust ports, as shown by Fig. 18. The steam ports are practically opposite each other, and the relation of the edges of the ports in the valve to these ports corresponds to that of the valve edges to the steam ports of the ordinary slide valve, forming the steam laps, lead and exhaust laps or clearance as the case may be. The exhaust cavity is a passage diametrically through the valve of sufficient width on one side to combine both steam ports with the main exhaust port simultaneously during the exhaust period. At right angles to the exhaust passage is a similar but somewhat larger cavity which corresponds to the steam chest with transverse passages through the valve body alternating with the exhaust passages, and the lap and exhaust edges are surrounded by carefully fitted slats, both on sides and ends to prevent leakage.

The motion is transmitted through a pivoted wrist plate to the valve from an ordinary Stephéson valve motion as indicated in Fig. 17. By means of pivoting the wrist plate on the arm of a bell crank whose other arm is connected with a union rod to a short arm on the reverse shaft, the wrist plate is raised and lowered by the motion of the reverse lever producing a moderate increase in lead, an earlier exhaust and later compression than the direct Stephenson motion produces in linking up the engine.

The main advantage of this valve is the quicker admission, closing and exhaust it accomplishes due to the double port openings, and the small resistance it offers to the valve motion, as compared with the slide valve, in being completely balanced. In common with the Allfree gear it gives a higher average pressure at high speeds than the ordinary valve and in a general way the illustration showing port openings in Fig. 16 is applicable also to this valve and gear. This gear, as well as in the previous case, involves additional complications over the ordinary gear, requiring special skill, both in its manufacture and adjustment, which to some extent counterbalances the above-named advantages.

The Use of Form 19 for Superior Trains.

BY H. A. DALBY.

At the recent convention of the Train Despatchers' Association of America, held at Buffalo, one of the interesting questions discussed was the use of the "19" order for all trains, superior as well as inferior. There were present members from single-track divisions of the Lake Shore and Pennsylvania, where this is the daily practice, and a number of other single-track lines are using it in this manner. With proper signal equipment at telegraph offices, and especially in conjunction with the "middle order," it has proved highly satisfactory, and the advantages gained over the usual method, that of requiring the conductor, and possibly the engineman, to go to the office and sign the 31 form, are certainly great.

There should be a special form of train-order signal, or an auxiliary to the regular signal, to indicate to the approaching train that a "19" order is to be delivered. The three-position signal has been suggested for this, using the caution color of the middle position of the arm for such indication, but this does not appear to commend itself as does the practice on the Lake Shore, where the signal is kept in the "stop" position and the operator goes to the platform and displays a white flag or light. On seeing this, the engineman reduces speed low enough so that the order may be delivered to himself and the conductor. A whistle signal would also be useful; the engineman could then notify the conductor to be prepared. This would be almost a necessity in the case of vestibuled passenger trains, on which some little time is required for a trainman to get to the steps.

The success of this use of the "19" order depends on the question of certainty in delivery. This is accomplished by the use of clearance cards under suitable regulations. The rule should be that when the train-order signal stands at "stop" no train may leave the station without a clearance card. This clearance card must show the number of each order delivered; or, if there are no orders, that fact must be stated. An additional feature of safety may be added by requiring that the operator check over with the despatcher the numbers of the orders to be delivered and receive the "O. K." or "Complete" to the card. The despatcher records this in his book and all concerned have a complete record. The train is not stopped, and the wire work consumes much less time than the process of getting and transmitting the conductor's signature.

The operator may be required to put down a torpedo as an additional stop signal, or a green flag or light may be used by him instead of the white. The "X" response may be applied to the 19 in the same way it is now used for the 31. Hoops may be provided in

steam pipes if necessary. The following table gives the principal dimensions of the three types:

Type engine & Gt. North. Class	Atlantic (K_4)	Pacific (H_2)	Prairie (J_1)
Weight on drivers	112,000 lbs.	151,000 lbs.	151,000 lbs.
" total of engine	211,000 "	227,000 "	209,000 "
Wheel base, driving	7 ft. 1½ in.	13 ft. 0 in.	13 ft. 0 in.
" total	30 " 4½ "	33 " 5½ in	30 " 9 in.
" engine and tender	62 " 10 in.	66 " 4 in.	63 " 7½ in.
Cylinders, diameter	11.75 in.	22 in.	22 in.
Piston stroke	26 in.	30 in.	30 in.
Valves	Piston.	Slide.	Slide.
Diameter of wheels, driving	73 in.	69 in.	69 in.
" " " eng. truck	36 in.	36 in.	36 in.
" " " trailing	49 in.	45 in.	45 in.
Boiler, diameter, front end	72 in.	72 in.	72 in.
Length of firebox	126¼ in.	126¼ in.	126¼ in.
Width of firebox	66¼ in.	66¼ in.	66¼ in.
Number of tubes	301	301	301
Diameter of tubes	2¼ in.	2¼ in.	2¼ in.
Length of tubes	18 ft. 6 in.	21 ft. 0 in.	18 ft. 6 in.
Working steam pressure	210 lbs.	210 lbs.	210 lbs.
Heating surface, tubes	3,277.89 sq. ft.	3,720.36 sq. ft.	3,277.89 sq. ft.
" " fire box	210.77 "	210.77 "	210.77 "
" " total	3,488.66 "	3,931.13 "	3,488.66 "
Journals, driving	10x10½ in.,	9½x12 in.	9½x12 in.
" eng. truck	6x12 in.	6x12 in.	6x12 in.
" trailers	8x12 in.	8x12 in.	8x12 in.
" tender	5½x10 in.	5½x10 in.	5½x12 in.
Tender capacity, water	8,000 gal.	8,000 gal.	8,000 gal.
Tender capacity, coal	13 tons.	13 tons.	13 tons.

Walschaert Valve Gear.*

The cost of maintenance of the Walschaert valve gear I believe to be fully 25 per cent. less than that of the Stephenson link motion. This estimate is based on figures obtained from the Rock Island roundhouse at Burr Oak, Ill., where there are a number of engines equipped with the Walschaert valve gear, while others are equipped with the Stephenson link motion, both classes being in freight service out of that point. Engine failures due to the breaking of valve motion connections during 12 months' service have been very slight. With 15 engines equipped with the Walschaert valve gear we have had only three failures that were chargeable to valve motion failing, while with engines equipped with the Stephenson link motion we have had at least three times as many failures due to the link motion getting out of order.

In view of the fact that the Walschaert link, not being influenced by a backing up eccentric, does not rock through as large an angle as the ordinary Stephenson link, this means that the link has a more favorable action on the link block with the Walschaert gear and does not have a tendency to spring the motion as much as the Stephenson link when at and near the extremity of its stroke. It should also be noticed that with the Walschaert gear all parts in connection with the valve move in almost a straight line, which does away with the up-and-down motion that is very noticeable with the Stephenson link. There is no question in my mind that this up-and-down motion has a tendency to cause a rocking action of the valve on its seat, whereas with the Walschaert link the valve has an easy gliding motion and is not so apt to hollow the valve seat.

My experience has been that where tonnage is considered the main factor and speed a secondary consideration, the Walschaert motion is superior to the Stephenson in tonnage capacity of engines, as it will handle a train better at a slow rate of speed with very little slipping, while with the Stephenson motion at a slow rate of speed the engine requires greater care in handling in order to prevent this trouble, making it necessary to use a greater amount of sand.

In regard to speed, I am not in a position to say which motion has the advantage, as none of our engines in passenger service are equipped with the Walschaert valve gear. In freight service we have about the same number of each, and they take their turn in all classes of freight, both fast and slow. It seems to be the general opinion of the men running these engines that engines equipped with the Stephenson motion can take a train of the same tonnage and make better time on the level or where the grade is in their favor than they can with those equipped with the Walschaert motion, but when they get on the hills or where the grade is against them they can make better time with the Walschaert than they can with the Stephenson motion. The only reason I can advance for this difference is the manner in which the valves are set, as both engines are of the same size and weight and carry the same steam pressure—185 lbs.

The valves on our engines equipped with the Stephenson link motion are set with 1-32 in. lead at full stroke, making about ¼ in. lead in the 6-in. cut-off; valve lap, 1 in. Valves on engines equipped with the Walschaert valve gear are set as follows: Five have ⅛ in. lead, which is constant, and 7⅛ in. lap; the remainder have 3-16 in. lead and 15-16 in. lap. These latter give the best results. It must be understood that the engines referred to were built for freight service and were not intended for high speed.

*Extracts from a paper presented to the convention of the Traveling Engineers' Association, by O. H. Rehmeyer, C., R. I. & P. Ry.

Unit Costs of Railroad Building.*

II.

The costs given below are for a first-class branch line 12 miles long, giving the St. Louis Southwestern an entrance into Dallas, Texas. The road is well ballasted and laid with 75-lb. rail.

(1) Length of new railroad, 12.13 miles main single track, 1.52 miles sidings.

(2) Year built, winter 1903-4.

(3) Cost for distance of 12.13 miles, including preliminary survey, clearing right of way, etc., but not including real estate, stations, equipment or signals, $19,649.24 per mile.

(4) Brief description: Line follows dividing ridge between two North Texas streams. No side hill cuts. Considerable fill but ground mostly level. Line being on ridge, but few openings required, all of which were made of a permanent character, including



Crossing of Dallas Branch of the St. Louis Southwestern over Missouri, Kansas & Texas Tracks.

concrete arch culverts, cast-iron pipe, vitrified tile drain, etc. Excavated material mostly earth; some loose rock.

The line is practically one tangent with the exception of 2,200 ft. of 1 deg. curve which occurs about midway and a few other light curves entering the suburbs of Dallas. The line crosses the M., K. & T. overhead on the edge of the Trinity river bottoms entering Dallas; the crossing consists of three 50-ft. deck plate girder spans carried on steel columns and concrete masonry abutments. Immediately after crossing the M., K. & T., the line enters the Trinity river bottoms and continues along the edge of the bottoms on a high embankment until the crossing of the Texas & Pacific on Pacific avenue, Dallas, is reached. The overhead crossing with the M., K. & T. required a heavy embankment through Trinity river bottoms, amounting to a quarter million cubic yards in a distance of about one mile. The material for this embankment was obtained by grading down ground to be used for an outer yard of the Dallas Terminal Railway and Union Depot Company. Steam shovels were used in excavating and the dirt hauled by train to make this fill. This method enabled the grading to be done for the outer yard, and at the same time gave waste material to fill the bottoms. The balance of the grading on the extension was done by teams.

The following table shows in detail the cost of the road:

Engineering	$6,322.57
Grading (of this amount $48,924.46 is chargeable to the Dallas extension proper, the balance to outer yard, Lamar street wye, etc.)	77,276.94
Bridges, trestles and culverts	41,661.75
Ties	26,838.38
Rails	37,607.06
Track fastenings	5,596.28
Frogs and switches	1,176.74
Ballast	30,526.14
Tracklaying and surfacing	9,510.71
Crossings, cattle guards and signs	1,928.81
Interlocking or signal apparatus	
Total	$238,345.18

*The first article of this series giving cost of railroad building in the Eastern States was printed in the *Railroad Gazette*, Sept. 7, 1906.